

EXPERIMENT 02

Recover Deleted or Damaged Files from a Storage Device Using TestDisk

OBJECTIVE

To recover deleted or damaged files from a storage device, analyze partition table structures, and restore lost data to a secure location using TestDisk.

TOOLS / REQUIREMENTS

- **Forensic Software:** TestDisk 7.3-WIP (Data Recovery Utility, Christophe GRENIER / CGSecurity)
- **Host Operating System:** Microsoft Windows 11
- **Storage Disk / Target Volume:** Virtual storage disk volume (\\.\PhysicalDrive1 — 2147 MB / 2048 MiB NTFS volume [New Volume])

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A storage volume was configured with a deleted test credential document (User_credentials.txt.txt). The forensic investigator was tasked with launching TestDisk, evaluating the partition table architecture, analyzing the volume structure, locating deleted file records from the Master File Table / directory index, and recovering the deleted artifact to a designated forensic directory without modifying the original evidence.

EVIDENCE / INPUT

- **Target Storage Volume:** Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - Msft Virtual Disk
- **Partition Structure:** 1 P HPFS - NTFS (CHS 261 255 63, Sector size 512, [New Volume])
- **Target Deleted Artifact:** ./User_credentials.txt.txt (Size: 18 bytes, Deleted Timestamp: 26-Aug-2026 19:29)
- **Recovery Destination Path:** C:\DFIR-Lab\

PROCEDURE

Step 1 — TestDisk Initialization and Log File Creation

The TestDisk executable (testdisk_win.exe) was launched with administrative privileges. The option [Create] Create a new log file was selected to record diagnostic messages and recovery actions in testdisk.log.

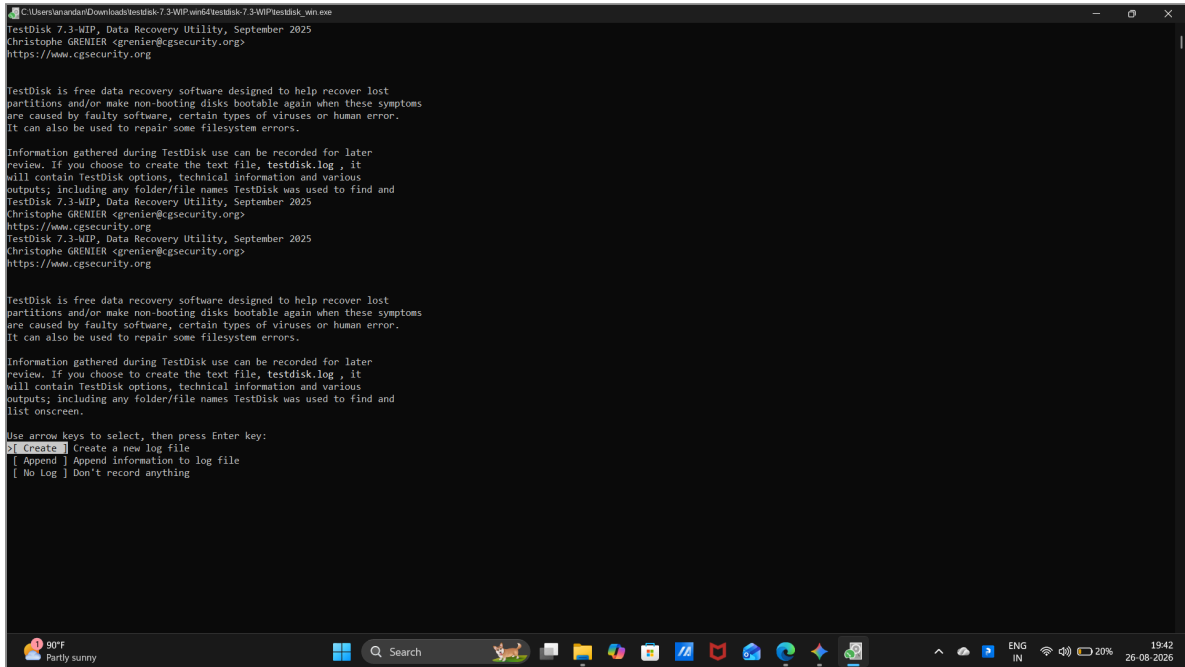


Figure 2.1: TestDisk startup and log file creation prompt

Step 2 — Storage Device and Partition Table Type Selection

The target virtual storage disk Disk \\.\PhysicalDrive1 (Msft Virtual Disk, 2048 MiB) was selected. The partition table type was set to [Intel] Intel/PC partition based on automated architecture detection.

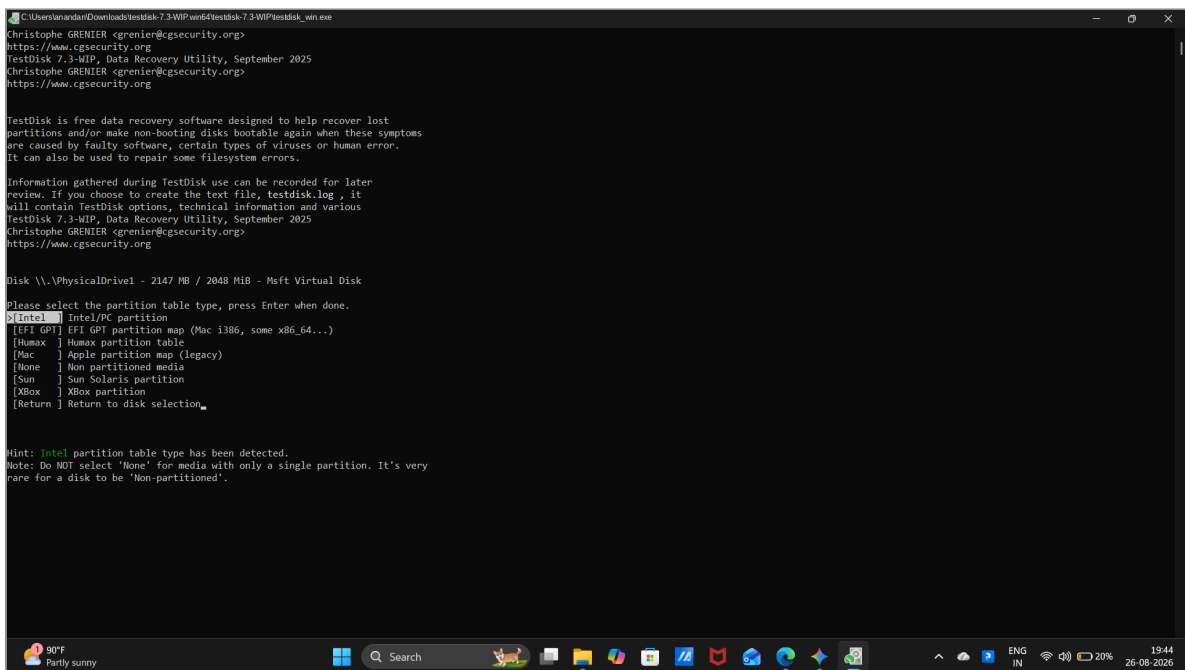


Figure 2.2: Storage device and Intel/PC partition table type selection

Step 3 — Current Partition Structure Analysis

The [Analyse] function was selected to inspect the volume geometry, check partition boundaries, and search for lost partitions.

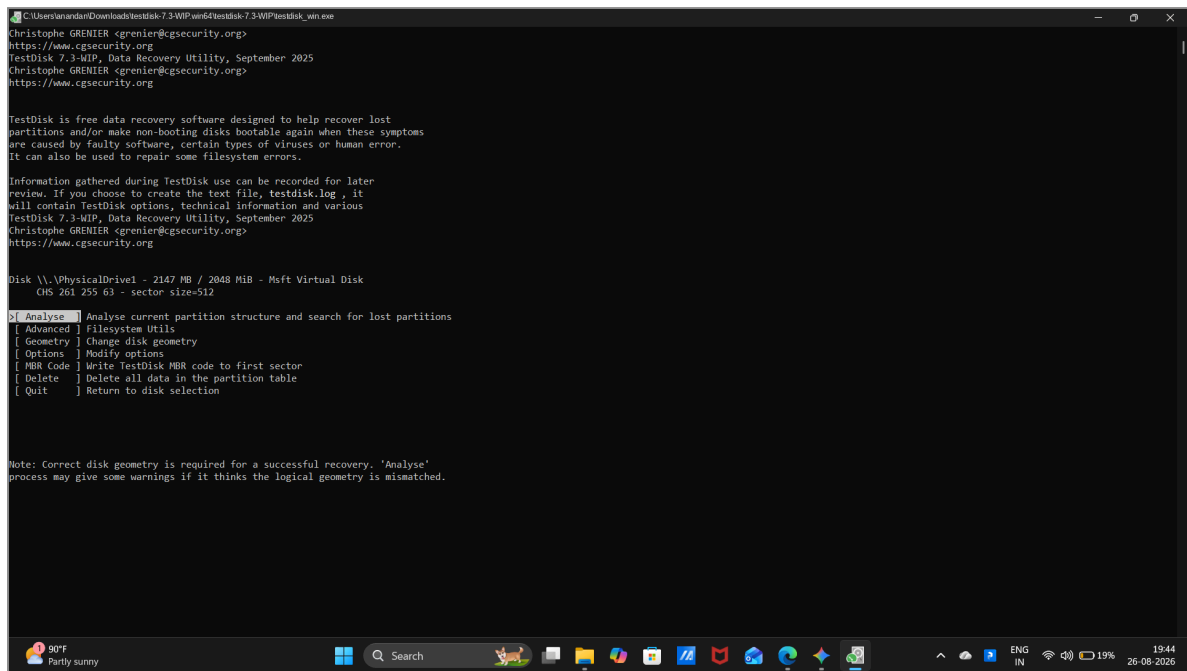


Figure 2.3: Current partition structure analysis

Step 4 — Navigating Advanced Filesystem Utilities

The [Advanced] Filesystem Utils menu was accessed to interact with filesystem-specific recovery mechanisms for the NTFS partition.

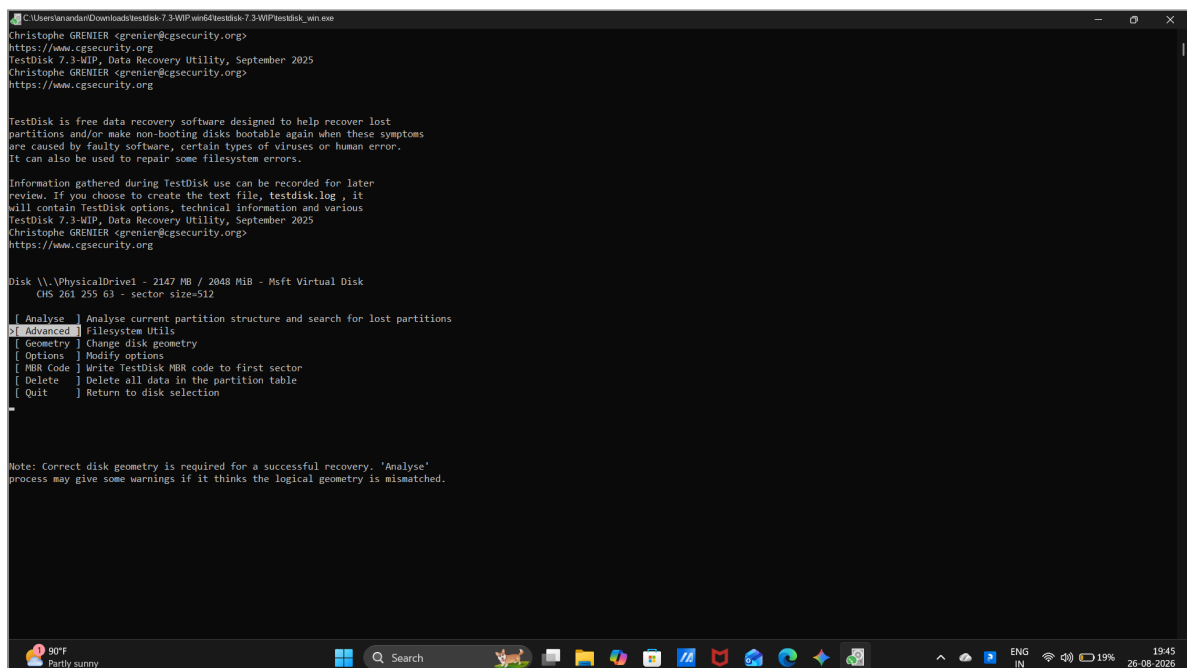


Figure 2.4: Advanced filesystem utilities menu

Step 5 — Inspecting Deleted Files on NTFS Volume

The undelete utility parsed the directory index and listed deleted file entries on [New Volume]. The deleted file `./User_credentials.txt.txt` (18 bytes, deleted 26-Aug-2026 19:29) was discovered.

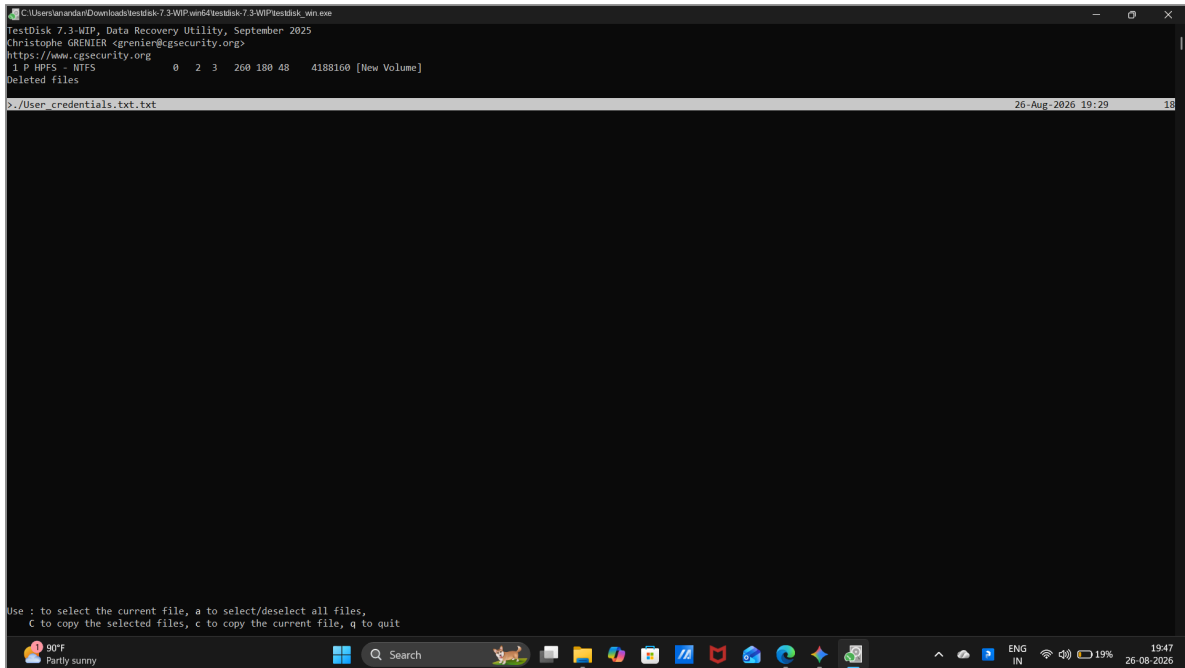


Figure 2.5: Deleted files listing on NTFS volume

Step 6 — Selecting File for Extraction and Recovery

The target artifact `./User_credentials.txt.txt` was selected and highlighted in green to prepare for copying out of unallocated space.

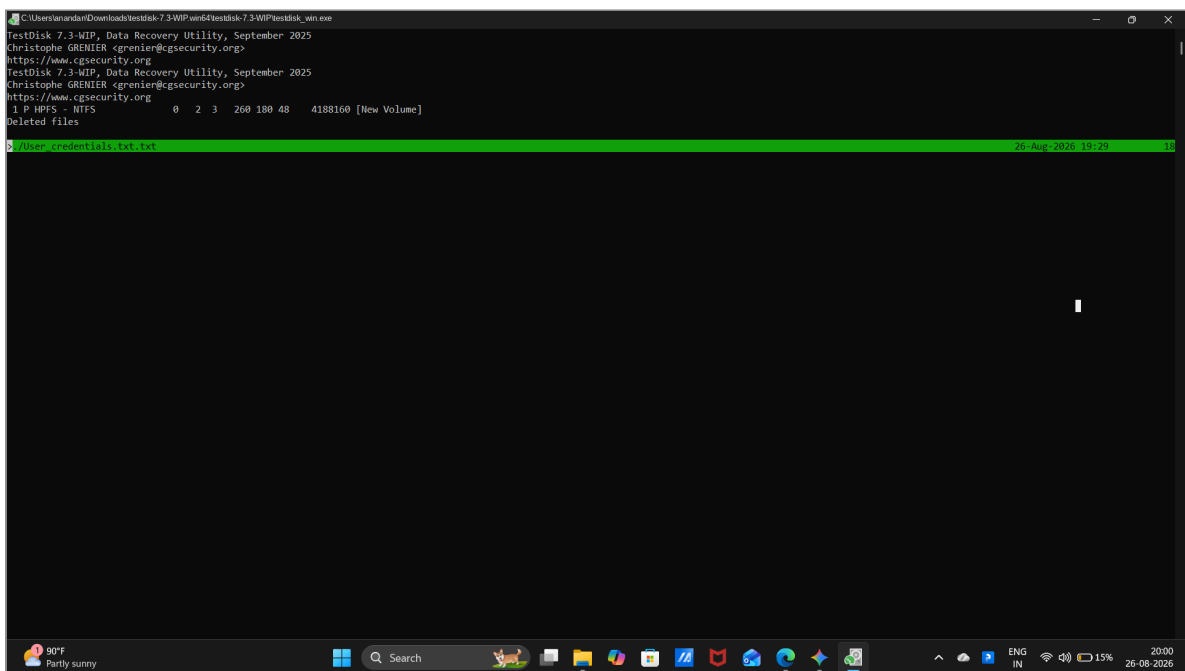


Figure 2.6: Selection and highlighting of deleted file for recovery

Step 7 — Choosing Secure Recovery Destination Directory

The destination drive path c was chosen to safely write out the recovered file outside the source storage volume.

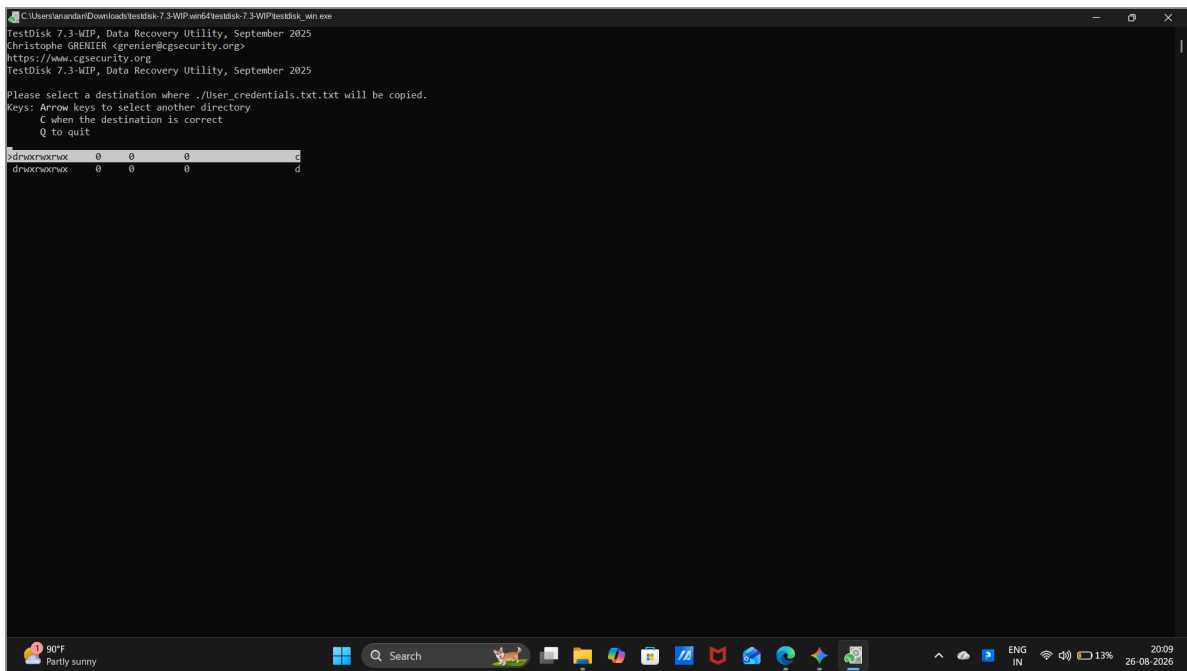


Figure 2.7: Selection of recovery destination directory

Step 8 — Verifying Recovered File in File Explorer

Windows File Explorer was opened to confirm successful restoration. The recovered file User_credentials.txt (1 KB, modified 26-08-2026 19:29) was verified inside C:\DFIR-Lab\.

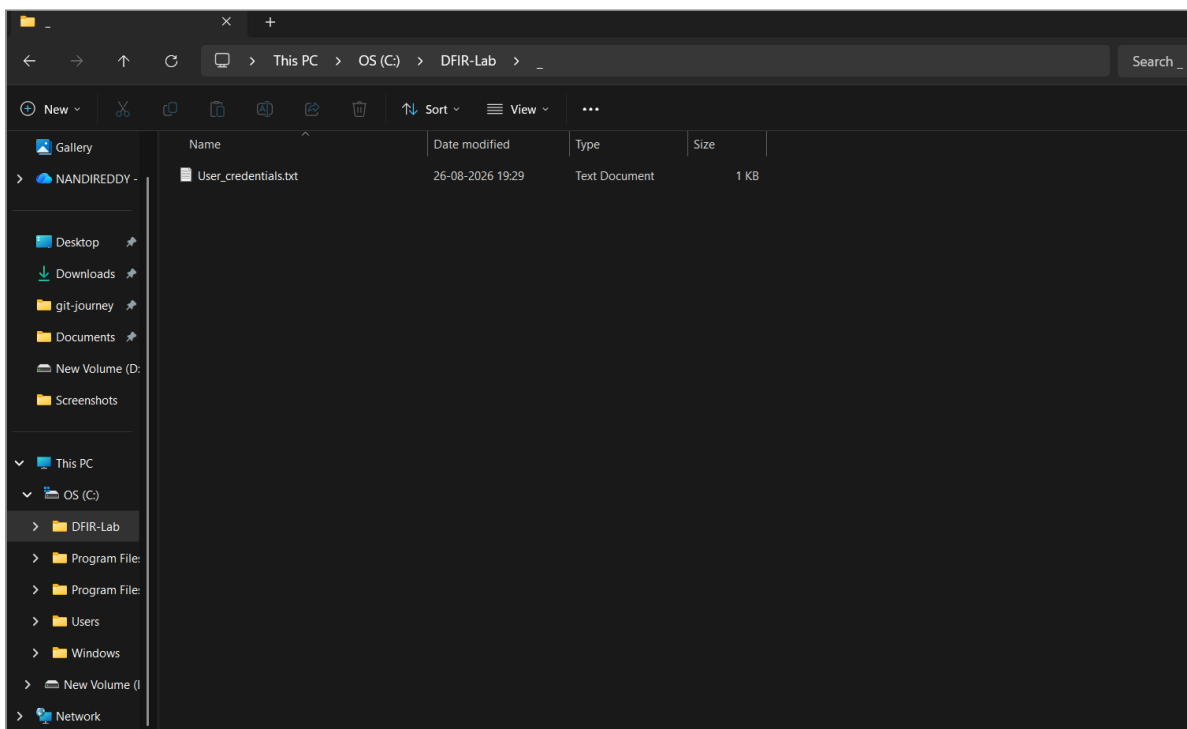


Figure 2.8: Verification of recovered file in Windows File Explorer

OBSERVATIONS

1. TestDisk accurately identified the virtual storage disk \\.\PhysicalDrive1 (2048 MiB).

2. The NTFS volume structure was recognized at CHS 261 255 63 ([New Volume]).
3. The deleted file entry `./User_credentials.txt.txt` was discovered with intact metadata (18 bytes).
4. The file was copied to `C:\DFIR-Lab\` and verified in File Explorer.

FINDINGS

- **MFT Record Preservation:** The NTFS file record remained intact in the Master File Table / directory index prior to cluster reallocation.
- **Forensic Recovery Integrity:** TestDisk successfully extracted and restored the deleted text file without altering the source volume structure.

RESULT

Successfully Completed. The deleted credential artifact was located on the NTFS storage volume, extracted, and successfully restored to `C:\DFIR-Lab\`.

CONCLUSION

TestDisk successfully identified the partition structure, scanned for deleted records, and recovered the lost text document to the forensic directory.

REFERENCES

- TestDisk Documentation (Christophe GRENIER / CGSecurity)
- Digital Forensics Laboratory Manual: Recover deleted or damaged files from a storage device using Test Disk